Distributed position estimation using *ARTVa* sensors during avalance *SAR* operations

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***Abstract*—Avalanche Search And Rescue operations (*SAR*) require a quick response time and efficient tools for locating avalanche victims, as the survival probability is closely related to burial time. The most widely used instrument is the *ARTVa* (*Apparecchio per la Ricerca di Travolti in VAlanga*); its effectiveness depends on the accuracy of the received signal and its correct usage, as well as many other factors.**
This paper explores how to improve coordination in *SAR* efforts when multiple rescuers are involved and improve the accuracy in estimating victims' locations. Unscented Kalman Filter (*UFK*), distributed consensus, Weighted Least Squares (*WLS*) and Weighted Geometric Dilution of Precision (*WGDOP*) concepts have been utilized to efficiently share and process information across different rescuers *ARTVa*s.
The proposed solutions have been validated in a *MATLAB* simulation environment.

I. Introduction

In recent years, participation in winter sports activities practiced in uncontrolled alpine environments has been steadily increasing. However, the very nature of these environments leads to high inherent danger: natural phenomena such as avalanches are often unpredictable, difficult to manage and very violent.
The main objective during avalanche *SAR* is the quick localization of buried victims. As shown in [1], survival chances are fairly high for any buried victim (provided that no other injuries have occurred) up to the 15-minute mark (92%); then the chances of life decreases drastically due to the effects of hypoxia and hypothermia.
An organized professional operation is often not fast enough as specialized rescuers are almost never on site; furthermore, cell signal and/or visibility might be greatly impaired. As a result, companion rescue remains the most effective option for rapid response. This is greatly aided by the use of avalanche transceivers or *ARTVa*s. Many Nations (such as Italy [2]) are mandating the use of such devices when engaging in alpine activities.
*ARTVa*s work by emitting a low pulsed signal at the reserved frequency of 457 kHz in transmission mode. The electromagnetic field generated by the victim's transmitter can be detected by rescuers' devices; signal intensity is used to estimate the buried victim position.
This approach is not without its drawbacks: signal quality can be affected by various factors including terrain configuration, general interference and the device instantaneous positioning and inclination. The advertised range of such devices ($\approx$ 70 m) is often misleading; in real-world conditions, as attested in [3] and [4], the real range can be considered to be around $\approx$ 30 m once the burial depth of the victim is accounted for.

A. Problem formulation

A perfectly executed standard *SAR* operation, according to [5] and [6], has the following steps:

1) *First signal search*: rescuers move systematically across the avalanche field following a structured search pattern called *Greca*: from their entry point, rescuers must follow a straight trajectory across the avalanche field while holding the *ARTVa* device in a horizontal position and maintaining a steady pace. After having traveled $d \approx 30$ m, a sharp is to be made. The movement is then repeated, turning in the opposite direction, creating a kind of *zig-zag* search pattern, ensuring that a large area can be covered without gaps. The turning points should be smooth and precise to maintain a consistent search width.
If the boundary of the search zone are reached, the rescuer must bounce off the border and continue.
2) *Coarse search*: once the first signal B_{victim} is found, the rescuer must move in the direction of the strongest signal while maintaining a controlled pace. The *ARTVa* should be rotated periodically to ensure the correct direction is being followed.
3) *Fine search*: as the detected magnetic field becomes more intense (and therefore the distance to the victim decreases), the rescuer must slow down and, while holding the *ARTVa* as close to the ground as possible, perform small lateral and forward movements to pinpoint the exact burial location.
4) *Pinpointing and digging phase*: once the strongest signal point is identified, a probe is used to confirm the burial depth of the victim before beginning excavation.

The different *SAR* phases can be visualized at Fig. 1.

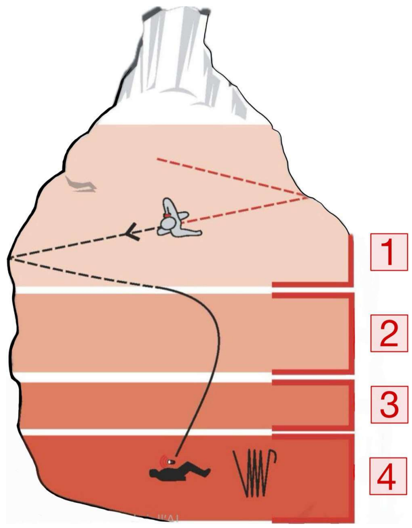

Fig. 1. Avalanche *SAR* standard operation phases [6]

Nowadays the market doesn't offer any option for *ARTVa* devices capable of communicating or sharing any kind of information with other transceivers: if multiple rescuers are involved, communication and coordination efforts are to be conducted using only visual and hearing cues. This is not ideal, especially during a high-stress-high-stakes situation where conditions are not ideal, line-of-sight communication might not be an option and leadership in operations isn't always straightforward.

In this study, *UKF*, consensus, *WLS* and *GDOP* have been applied to the *ARTVa* system to enhance coordination and improve the estimation of the avalanche victim's location. By enabling information sharing between multiple rescuers, these methods refine position estimates and reduce uncertainty in the search process.
To support the transceivers in position estimation, *UWB* sensors have been integrated: their range capabilities, compact size and robustness make them well-suited for avalanche rescue scenarios [7]. Additionally, their lower cost compared to high-precision alternatives like *GPS* (whose signal may be unreliable due to poor satellite coverage in mountainous or obstructed environments) makes them a practical and scalable solution.

The proposed revision to the *SAR* operation will be the following:

1) The *first signal search* remains unchanged. Each rescuer is equipped with two *UWB* sensors on one of the skies, so that he can estimate all the other rescuers' positions.
2) Once the first rescuer intercepts the victim signal B_{victim}, all the others converge very quickly to its estimated position. In the meantime, he has begun to follow the magnetic field intensity lines and getting closer to the victim.
   If, before reaching the first rescuer, the others also pick up the victim signal, they also follow the *coarse search* standard protocol.
3) Once at least 3 rescuers have a good lock on the magnetic field distance, the victim position estimation can be performed and dynamically adjusted. This happens before the standard *fine search* phase, allowing for greater time savings and positional accuracy.
4) The *pinpointing and digging phase* is not modified. However, given that the estimated victim position has been defined during step 3, the need for probing is removed and this step becomes faster.

II. Model description

A. Working area

Avalanches start from a slope with an inclination $> 30°$ [3] and slide down towards the valley floor. Once settled, large patches are covered. Of course, actual dimensions are highly dependent on temperature, relative inclination of the slope, the presence of trees and other obstacles, humidity, cohesion between different snow substrates, and many other factors. For this study, a medium-sized avalanche has been simulated considering a flat $2D$ square patch of $200\ m \times 200\ m$. This definition can be considered without loss of generality, as inclination is only relevant to the walking speed of the rescuers, the burial depth of the victim has been accounted for in the distance at which the *ARTVa* signal can be picked up by any rescuers, and the presence of any natural obstacles that may impair movement or signal detection is case-specific.

B. Buried victim

The victim has been modeled as a point placed randomly inside the limits of the working area. Its orientation, once again defined randomly, determines the overall angle of the unitary magnetic dipole m whose magnetic field B simulates the transmitting *ARTVa* signal [8]. In the $2D$ working scenario, the following equations hold true:

$$m = [m_x, m_y] = [\cos(\theta), \sin(\theta)] \tag{1}$$

$$B = \frac{1}{r^4}(3(m \cdot r)r - mr^2) \Rightarrow \begin{cases} B_x = \frac{3x(m_x x + m_y y) - m_x r^2}{r^4} \\ B_y = \frac{3y(m_x x + m_y y) - m_y r^2}{r^4} \end{cases} \tag{2}$$

$$r = \sqrt{(x_0 - x)^2 + (y_0 - y)^2} \tag{3}$$

where r is the distance between the victim (located at $\{x_0, y_0\}$) and the rescuer (located at $\{x, y\}$).
Equations (1) and (2) define B throughout the working space. However, to establish the threshold for the first detectable signal, an equipotential contour of $|B|$ is defined at a $d \approx 30\ m$ radius around the victim. Additionally, a second smaller equipotential line marks the boundary where the victim's position estimation becomes feasible.
The high operating frequency of the device allows the orientation of the B field to be disregarded. As a result, during the *coarse search* phase, rescuers follow the path of maximum field magnitude without considering its $sign$. Consequently, the victim's magnetic field can be represented using contour lines rather than oriented vectors.

C. Rescuers

Each rescuer has been defined using a simple discrete-time unicycle vehicle model [9]. The state of each rescuer is fully defined by a three-element column vector:

$$X = \begin{bmatrix} x & y & \theta \end{bmatrix}^\top = \begin{bmatrix} x_1 & x_2 & \theta \end{bmatrix}^\top \tag{4}$$

where x, y are the Cartesian coordinates and θ represents the orientation angle with respect to the global reference frame. The evolution of the kinematic model at step $k+1$ can be defined as follows:

$$X_{k+1} = f(X_k) = \begin{bmatrix} x_{1,k+1} \\ x_{2,k+1} \\ \theta_{k+1} \end{bmatrix} = \begin{bmatrix} x_{1,k} + v_k \cos(\theta_k) dT \\ x_{2,k} + v_k \sin(\theta_k) dT \\ \theta_k + \omega_k dT \end{bmatrix} \tag{5}$$

where:

- v_k is the linear velocity of the rescuer at time step k.
- ω_k is the angular velocity of the rescuer at time step k.
- dT is the time step of the system. Accounting for the *UWB*, a working frequency of $10\ Hz$ has been considered.

Using this model, rescuers can be represented using oriented triangles, as seen in Fig. 2.
A worst-case scenario has been assumed: none of the rescuers are able to see or communicate with one another; the only information at their disposal is the estimated data coming from the aforementioned sensors.

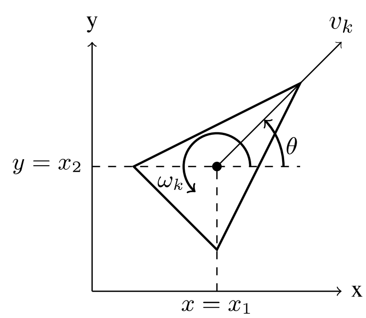

Fig. 2. Rescuers' unicycle model

D. Rescuers' movement and control

The goal of the rescuers is to adhere as much as possible to the movements described in (I-A). However, some human behavior imperfections must be taken into account: random variability about the traveled distance, walking speed and turning angle have been introduced. Additionally, rescuers' trajectories are not perfectly straight: small angular drift allows for a better representation of real-world movements.

$$d = d_{\text{base}} + \mathcal{N}(0, \sigma_d^2) \tag{6}$$

$$\theta = \theta_{\text{base}} + \mathcal{N}(0, \sigma_\theta^2) \tag{7}$$

$$v = v_{\text{base}} + \mathcal{N}(0, \sigma_v^2) \tag{8}$$

Once the first rescuer enters the *ARTVa* detection zone, its movement is adjusted to maximize search efficiency, following the gradient of the magnetic field. Speed and direction control schemes are regulated based on the intensity of the received signal.
The magnetic field components $[B_x, B_y]$ at the rescuer's estimated position $\{x_{1,\text{first}}, x_{2,\text{first}}\}$ are calculated using (1), (2) and (3). Its heading angle towards the victim is estimated as:

$$\theta_{\text{victim}} = \arctan\left(\frac{B_y}{B_x}\right) + \mathcal{N}(0, \sigma^2_{\theta_{\text{ARTVa}}}) \tag{9}$$

where $\sigma_{\theta_{\text{ARTVa}}}$ represents the uncertainty in the heading measurement due to the *ARTVa* sensor.
Given the rescuer orientation θ_{first}, the angular error $e_{\theta,\text{first}}$ is defined as:

$$e_{\theta,\text{first}} = \theta_{\text{victim}} - \theta_{\text{first}} \tag{10}$$

To align the rescuer's heading, its angular velocity ω_i is updated using a proportional control law:

$$\omega_{\text{first}} = k_\omega \ \arctan\left(\frac{\sin(e_{\theta,\text{first}})}{\cos(e_{\theta,\text{first}})}\right) \tag{11}$$

Rescuers who have not yet entered the *ARTVa* detection zone navigate towards the estimated position of the first rescuer using a control law that minimizes their distance while aligning their heading.

- *Angular velocity control*: denoting as $\{x_{1,i}, x_{2,i}\}$ the estimated position of each rescuer outside the detection zone, the error in their position relative to the first rescuer is defined as:

$$\begin{aligned} e_{x,i} &= x_{1,\text{first}} - x_{1,i} \\ e_{y,i} &= x_{2,\text{first}} - x_{2,i} \end{aligned} \tag{12}$$

  The desired heading direction toward the first rescuer is given by:

$$\theta_{\text{first}} = \arctan\left(\frac{e_{y,i}}{e_{x,i}}\right) \tag{13}$$

  The heading error is computed as:

$$e_{\theta_i} = \theta_{\text{first}} - \theta_i \tag{14}$$

  This error is corrected by adjusting the rescuers' angular velocity ω_i using a proportional control law:

$$\omega_i = k_\omega \ e_{\theta_i} \tag{15}$$

- *Linear velocity control*: The speed at which each rescuer converges towards the first rescuer is proportional to their distance. This is enforced thorough a proportional control

$$v_i = k_v \ \sqrt{e^2_{x,i} + e^2_{y,i}} \tag{16}$$

All rescuers stop once they get very close to the victim ($d_{\text{ref}} \approx 1\ m$ radius), as the detected magnetic field strength exceeds a set threshold. This simulates the beginning of step 4 of the *SAR* operation, which has not been modeled.

III. Rescuers' positions estimation algorithm

Rescuers' relative position estimation inside the working area has been determined using an algorithm that combines trilateration, *UKF* and distributed weighted consensus approaches.

A. Preliminary steps

UWB sensors are employed to estimate the distances between rescuers. By mounting them on skis at fixed, known coordinates, they also serve as reference anchors, following a circular indexing logic:

- n rescuers have been defined, each identifiable by an index $i \in [1, n]$.
- Each one has 2 *UWB* anchors placed symmetrically at $a_1 = \begin{bmatrix} x_i & (y_i + d_{\text{anc}}) \end{bmatrix}^\top$ and $a_2 = \begin{bmatrix} x_i & (y_i - d_{\text{anc}}) \end{bmatrix}^\top$, according to the rescuer internal reference frame (Fig. 3).
- Each rescuer i uses neighboring rescuers $i-1$ and $i+1$ to define a first set of 4 reference anchors (17). Indexing is circular: if $i+1 > n$, then $i+1 = 1$; if $i-1 < 1$, then $i-1 = n$.
- A second set of 4 anchors (18) is constructed for each rescuer i using neighbors $i-1$ and $i+2$, again following a circular indexing logic.

$$\begin{aligned}\text{anc}_{i,1} &= \begin{bmatrix} x_{i-1} & x_{i-1} & x_{i+1} & x_{i+1} \\ y_{i-1} + d_{\text{anc}} & y_{i-1} - d_{\text{anc}} & y_{i+1} + d_{\text{anc}} & y_{i+1} - d_{\text{anc}} \end{bmatrix} = \\ &= \begin{bmatrix} a_{1,1} & a_{2,1} & a_{3,1} & a_{4,1} \end{bmatrix}\end{aligned} \tag{17}$$

$$\begin{aligned}\text{anc}_{i,2} &= \begin{bmatrix} x_{i-1} & x_{i-1} & x_{i+2} & x_{i+2} \\ y_{i-1} + d_{\text{anc}} & y_{i-1} - d_{\text{anc}} & y_{i+2} + d_{\text{anc}} & y_{i+2} - d_{\text{anc}} \end{bmatrix} = \\ &= \begin{bmatrix} a_{1,2} & a_{2,2} & a_{3,2} & a_{4,2} \end{bmatrix}\end{aligned} \tag{18}$$

The 2 sets of anchors are to be considered as separate ($j = 1$ if $\text{anc}_{i,1}$, $j = 2$ if $\text{anc}_{i,2}$). For each of them, the Euclidean distance between the set's 4 anchors $a_{m,j}$ and both i^{th}-rescuer *UWB*s (a_1 and a_2) can be defined. For example, taking $j = 1$, results are the following:

$$\begin{aligned}\text{dist}_{1,i}^{(m)} &= \|a_{k,1} - a_1\| = \\ &= \sqrt{\left(x_{\text{anc}_{i,1}} - x_i\right)^2 + \left(y_{\text{anc}_{i,1}} - (y_i + d_{\text{anc}})\right)^2} = \\ &= \begin{bmatrix} \sqrt{(x_{i-1} - x_i)^2 + (y_{i-1} + d_{\text{anc}} - (y_i + d_{\text{anc}}))^2} \\ \sqrt{(x_{i-1} - x_i)^2 + (y_{i-1} - d_{\text{anc}} - (y_i + d_{\text{anc}}))^2} \\ \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} + d_{\text{anc}} - (y_i + d_{\text{anc}}))^2} \\ \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - d_{\text{anc}} - (y_i + d_{\text{anc}}))^2} \end{bmatrix} = \\ &= \begin{bmatrix} \sqrt{(x_{i-1} - x_i)^2 + (y_{i-1} - y_i)^2} \\ \sqrt{(x_{i-1} - x_i)^2 + (y_{i-1} - y_i - 2d_{\text{anc}})^2} \\ \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2} \\ \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i - 2d_{\text{anc}})^2} \end{bmatrix} = \begin{bmatrix} \text{dist}_{1,i}^{(1)} \\ \text{dist}_{1,i}^{(2)} \\ \text{dist}_{1,i}^{(3)} \\ \text{dist}_{1,i}^{(4)} \end{bmatrix}\end{aligned} \tag{19}$$

$$\begin{aligned}\text{dist}_{2,i}^{(m)} &= \|a_{k,1} - a_2\| = \\ &= \sqrt{\left(x_{\text{anc}_{i,1}} - x_i\right)^2 + \left(y_{\text{anc}_{i,1}} - (y_i - d_{\text{anc}})\right)^2} = \\ &= \begin{bmatrix} \sqrt{(x_{i-1} - x_i)^2 + (y_{i-1} + d_{\text{anc}} - (y_i - d_{\text{anc}}))^2} \\ \sqrt{(x_{i-1} - x_i)^2 + (y_{i-1} - d_{\text{anc}} - (y_i - d_{\text{anc}}))^2} \\ \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} + d_{\text{anc}} - (y_i - d_{\text{anc}}))^2} \\ \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - d_{\text{anc}} - (y_i - d_{\text{anc}}))^2} \end{bmatrix} = \\ &= \begin{bmatrix} \sqrt{(x_{i-1} - x_i)^2 + (y_{i-1} - y_i + 2d_{\text{anc}})^2} \\ \sqrt{(x_{i-1} - x_i)^2 + (y_{i-1} - y_i)^2} \\ \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i + 2d_{\text{anc}})^2} \\ \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2} \end{bmatrix} = \begin{bmatrix} \text{dist}_{2,i}^{(1)} \\ \text{dist}_{2,i}^{(2)} \\ \text{dist}_{2,i}^{(3)} \\ \text{dist}_{2,i}^{(4)} \end{bmatrix}\end{aligned} \tag{20}$$

Similar considerations apply for $j = 2$.

Results (19) and (20) are to be averaged together to obtain a single estimated distance between rescuer i and each of the $m \in [1, 4]$ belonging to its neighboring rescuers, obtaining (21). This result has been used to estimate through trilateration the position of each rescuer.

$$\text{dist}_i = \frac{1}{2}\left(\text{dist}_{1,i}^{(m)} + \text{dist}_{2,i}^{(m)}\right) \Rightarrow \text{dist}_i = \begin{bmatrix} \text{dist}_i^{(1)} \\ \text{dist}_i^{(2)} \\ \text{dist}_i^{(3)} \\ \text{dist}_i^{(4)} \end{bmatrix} \tag{21}$$

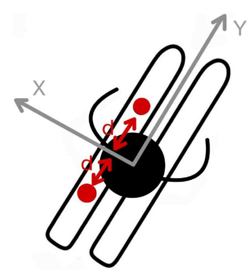

Fig. 3. Sketch of the *UWB* configuration for each rescuer, top view

B. Rescuers' trilateration algorithm

The goal of the trilateration is to estimate the position of each rescuer in the working area using the distance measurements and accounting for range uncertainties.

Each distance measurement $\text{dist}_i^{(k)}$ defines the radius of a circle centered in the k^{th} anchor. The rescuer's estimated position lies at the intersection of these circles. By looping the algorithm around all n rescuers, all their relative positions are obtained.

Considering $[x, y]$ as the unknown position to be determined of the rescuer and $[x_i, y_i]$ as the position of the k^{th} anchor, each distance measurement can be squared:

$$\begin{aligned}\left(\text{dist}_i^{(m)}\right)^2 &= \text{dist}_i^2 = (x_i - x)^2 + (y_i - y)^2 = \\ &= x_i^2 + x^2 - 2x_i x + y_i^2 + y^2 - 2y_i y\end{aligned} \tag{22}$$

From an algebraic point of view, this leads to a nonlinear system of equations. It can be solved by applying a Least squares algorithm (*LS*): by subtracting two distances with respect to consecutive anchors, the quadratic unknown terms (x^2 and y^2) are canceled out, obtaining a linearized problem.

$$
\begin{aligned}
\text{dist}_{i+1}^2 - \text{dist}_i^2 &= (x_{i+1} - x)^2 + (y_{i+1} - y)^2 - \\
&- (x_i - x)^2 - (y_i - y)^2 = \\
&= x_{i+1}^2 + x^2 - 2x_{i+1}x + y_{i+1}^2 + y^2 - \\
&- 2y_{i+1}y - x_i^2 - x^2 + 2x_ix - \\
&- y_i^2 - y^2 + 2y_iy = \\
&= x_{i+1}^2 - x_i^2 + y_{i+1}^2 - y_i^2 - \\
&- 2(x_{i+1} - x_i)x - 2(y_{i+1} - y_i)y
\end{aligned} \tag{23}
$$

Expanding the equations to all possible combinations of rescuer and anchors, equation (23) can be summed up using matrices. For each rescuer i, the system can be rewritten as:

$$H \cdot x = z \tag{24}$$

where:

- n is the number of anchors used for each rescuer.
- $x = [x \quad y]^\top \in \mathbb{R}^{2\times1}$ is the vector containing each rescuer position
- $z \in \mathbb{R}^{(n-1)\times1}$ is the vector representing the difference in squared distances between consecutive anchors
$$z_i = \text{dist}_i^2 - \text{dist}_{i+1}^2 + x_{i+1}^2 - x_i^2 + y_{i+1}^2 - y_i^2 \tag{25}$$
- Matrix $H \in \mathbb{R}^{(n-1)\times2}$ represents the difference between two consecutive anchor positions. Each row can be computed as:
$$H_i = 2 \cdot \begin{bmatrix} x_{i+1} - x_i & y_{i+1} - y_i \end{bmatrix} \tag{26}$$

Uncertainty in each measurement is also to be considered. Each rescuer's position is the result of a trilateration process based on 4 anchors, whose positions are themselves affected by estimation errors, as every rescuer simultaneously serves as both an anchor and an unknown for the others (as described in Section (III-A)). As a result, measurement uncertainty propagates from anchor position estimation to distance estimation. To account for this, a covariance matrix $C \in \mathbb{R}^{(n-1)\times(n-1)}$ has been defined according to the *propagation of errors* principles.

The covariance matrix C is defined as:

$$C = J \cdot C_{\text{dist}} \cdot J^\top \tag{27}$$

where:

- C_{dist} represents the noise contribution σ_i^2 of each new distance measurement dist_i, assuming it to be Gaussian and *i.i.d.*
$$C_{\text{dist}} = \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n^2 \end{bmatrix} \tag{28}$$
In this context all variances have been considered to be equal, so $\sigma_1^2 = \dots = \sigma_i^2 = \dots = \sigma_n^2$.
- $J \in \mathbb{R}^{(n-1)\times n}$ represents the Jacobian matrix of z_i with respect to distances dist_i
$$J_{i,j} = \frac{\partial z_i}{\partial \text{dist}_j} \tag{29}$$
According to the structure of z_i described in (25), only 2 terms in each row i of J are non-zero:
$$
\begin{aligned}
J_{i,i} &= \frac{\partial z_i}{\partial \text{dist}_i} = 2 \cdot \text{dist}_i \\
J_{i,i+1} &= \frac{\partial z_i}{\partial \text{dist}_{i+1}} = -2 \cdot \text{dist}_{i+1}
\end{aligned} \tag{30}
$$

Compiling together (27), (28), (29) and (30), the non-zero elements of matrix C have been computed as:

$$
\begin{aligned}
C_{i,i} &= J_{i,i}^2 \cdot \sigma_i^2 + J_{i,i+1} \cdot \sigma_{i+1}^2 = \\
&= (-2 \cdot \text{dist}_{i+1})^2 \cdot \sigma_{i+1}^2 + (2 \cdot \text{dist}_i)^2 \cdot \sigma_i^2 = \\
&= 4\sigma_{i+1}^2 \cdot \text{dist}_{i+1}^2 + 4\sigma_i^2 \cdot \text{dist}_i^2
\end{aligned} \tag{31}
$$

$$
\begin{aligned}
C_{i,i+1} = C_{i+1,i} &= J_{i,i+1} \cdot J_{i+1,i+1} \cdot \sigma_{i+1}^2 = \\
&= (-2 \cdot \text{dist}_{i+1})(2 \cdot \text{dist}_{i+1}) \cdot \sigma_{i+1}^2 = \\
&= -4\sigma_{i+1}^2 \cdot \text{dist}_{i+1}^2
\end{aligned} \tag{32}
$$

Matrices H, C and z have then been passed to the *UKF* algorithm to refine the position estimate of each rescuer, integrating the measurement information from trilateration with the predicted position in a recursive manner.

C. UKF algorithm

Given the nonlinear nature of the system described in (5), an Unscented Kalman Filter (*UKF*) has been utilized to estimate the position and orientation of each rescuer by recursively combining motion predictions and trilateration measurements. Unlike the Extended Kalman Filter (*EKF*), which relies on local linearization of the system dynamics, the *UKF* uses sigma points to better capture the nonlinearities of the system. These sigma points are selected to represent the mean and covariance of the current state estimate and are propagated through the nonlinear process model. This leads to a better approximation of the true state distribution.

The prediction is then refined using the actual measurements coming from the trilateration during the correction step, resulting in improved estimation accuracy under nonlinear dynamics.

According to [9] and [10], the steps of the *UKF* algorithm are the following:

1) *Sigma point definition*

Given $L = 3$ as the number of state variables of the system (4), $2L + 1$ sigma points are defined for the current estimate. Considering:

- $\hat{x}_k \in \mathbb{R}^L$ as the state estimate at the k^{th} time step
- $P_k \in \mathbb{R}^{L\times L}$ as its associated covariance matrix
- $\lambda = \alpha^2 (L + \kappa) - L$ as the *UKF* scaling factor (where α and κ are tuning parameters [9])
- $m = 0, \dots, 2L$ as the indexing variable

the sigma points at time step k have been defined as:

$$\chi_k^{[m]} = \begin{cases} \hat{x}_k & \text{if } m = 0 \\ \hat{x}_k + \sqrt{L+\lambda} \cdot \left[\sqrt{P_k}\right]_m & \text{if } m = 1, \dots, L \\ \hat{x}_k - \sqrt{L+\lambda} \cdot \left[\sqrt{P_k}\right]_{m-L} & \text{if } m = L+1, \dots, 2L \end{cases} \tag{33}$$

where $\left[\sqrt{P_k}\right]_m$ is the m^{th} column of the matrix square root of P_k, which has been computed numerically using Cholesky

decomposition.
Each sigma point $\chi_k^{[m]}$ is transformed through the process prediction model $f(\cdot)$ (which is the nonlinear system dynamics function described at (5)), thus obtaining the predicted sigma points $\chi_{k+1|k}^{[m]}$.

$$\chi_{k+1|k}^{[m]} = f\left(\chi_k^{[m]}\right) \tag{34}$$

These are then used to compute the predicted state mean and covariance.

2) *Prediction step*

This step computes the prior estimate of the state mean (predicted state) and the predicted state covariance as follows:

$$\hat{x}_{k+1|k}^{-} = \sum_{m=0}^{2L} W_m^{(m)} \cdot \chi_{k+1|k}^{[m]} \tag{35}$$

$$P_{k+1|k}^{-} = \sum_{m=0}^{2L} W_m^{(c)} \cdot \left(\chi_{k+1|k}^{[m]} - \hat{x}_{k+1|k}\right)\left(\chi_{k+1|k}^{[m]} - \hat{x}_{k+1|k}\right)^{\top} + Q_k \tag{36}$$

where:

- $W_m^{(m)}$ represent the mean weights associated with each sigma point

$$W_m^{(m)} = \begin{cases} \frac{\lambda}{\lambda+L} & \text{if } m = 0 \\ \frac{1}{2(\lambda+L)} & \text{if } m = 1, \ldots, 2L \end{cases} \tag{37}$$

- $W_m^{(c)}$ represent the covariance weights associated with each sigma point (given β as a tuning parameter [9])

$$W_m^{(c)} = \begin{cases} W_0^{(m)} + \left(1 - \alpha^2 + \beta\right) & \text{if } m = 0 \\ W_m^{(m)} & \text{if } m = 1, \ldots, 2L \end{cases} \tag{38}$$

- Q defines the process noise covariance matrix, which represents the uncertainty introduced by the system dynamics. For each rescuer i, it is defined as:

$$Q_{i,k} = \begin{bmatrix} (\sigma_v \cdot \cos(\theta_k))^2 & 0 & 0 \\ 0 & (\sigma_v \cdot \sin(\theta_k))^2 & 0 \\ 0 & 0 & \sigma_\omega^2 \end{bmatrix} \tag{39}$$

given σ_v as the uncertainty on the speed (defined in (8)) and σ_ω as the uncertainty on the angular velocity.

The resulting predicted state and covariance are used in the measurement update step.

3) *Measurement update step*

Given the prediction on the rescuers' positions, the *UKF* performs a correction by incorporating the actual measurements coming from the trilateration algorithm (Section III-B). each sigma-point is recalculated using the system measurement function $h(\cdot)$, which maps the state vector (4) to the measured variables. This step adjusts the estimated state based on the observed measurements (rather than propagating the state over time as in the *prediction step*).

$$z_k^{[m]} = h\left(\chi_{k+1|k}^{[m]}\right) \tag{40}$$

Then, the filter innovation is computed, consisting of the estimated measurement mean $\hat{z}_k$ (predicted measurement) and the innovation covariance matrix S:

$$\hat{z}_k = \sum_{m=0}^{2L} W_m^{(m)} z_k^{[m]} \tag{41}$$

$$S = \sum_{m=0}^{2L} W_m^{(c)} \left(z_k^{[m]} - \hat{z}_k\right)\left(z_k^{[m]} - \hat{z}_k\right)^{\top} + C \tag{42}$$

C is the covariance matrix coming from the trilateration measurements defined in (27), representing the uncertainty in the distance-based measurements.
Consequently, the cross-covariance matrix is calculated as:

$$P_{xz} = \sum_{m=0}^{2L} W_m^{(c)} \left(\chi_{k+1|k}^{[m]} - \hat{x}_{k+1|k}\right)\left(z_k^{[m]} - \hat{z}_k\right)^{\top} \tag{43}$$

This matrix captures the relationship between the predicted state and the expected measurement errors.
The Kalman gain of the filter, which determines how much the measurements influence the state update, is defined as:

$$K = P_{xz} S^{-1} \tag{44}$$

The posterior state estimate and covariance are then computed as:

$$\hat{x}_{k+1|k+1} = \hat{x}_{k+1|k}^{-} + K\left(z_k - \hat{z}_k\right) \tag{45}$$

$$P_{k+1|k+1} = P_{k+1|k}^{-} - KSK^{\top} \tag{46}$$

where:

- z_k is the actual measurement from the trilateration step at time step k described in (25).
- $P_{k+1|k+1}$ is the updated state covariance after measurement incorporation.

D. Consensus algorithm

The last step of the position estimation algorithm involves refining each rescuer's state estimate through a distributed weighted average consensus process [10].
At each time step k, $n_{\text{est}} = 2$ separate estimates $\hat{x}_{i,j}$ and their corresponding covariances $P_{i,j}$ are generated for each rescuer i using the two different anchor configurations $\text{anc}_{i,j}$ (where $j \in [1, 2]$, defined in equations (17) and (18)).
Consensus aligns these two estimates using weights. At each consensus iteration ν, the state and covariance are updated as:

$$\hat{x}_i^{(\nu+1)} = \sum_{j=1}^{2} \pi_{ij} \hat{x}_j^{(\nu)} \tag{47}$$

$$P_i^{(\nu+1)} = \sum_{j=1}^{2} \pi_{ij} P_j^{(\nu)} \tag{48}$$

where:

- π_{ij} are the consensus weights, computed according to the Metropolis-Hastings algorithm:

$$\pi_{ij} = \begin{cases} \frac{1}{\max\left(\phi_i, \phi_j\right)+1} & \text{if } j \in [1,2], j \neq i \\ 1 - \sum_{j \in [1,2], j \neq i} \pi_{ij} & \text{if } i = j \\ 0 & \text{otherwise} \end{cases} \tag{49}$$

- ϕ_i (ϕ_j) is the degree of node i (j).
  This quantity denotes the number of neighbors that can send and receive the information to and from node i (j). In this context, each rescuer generates 2 internal estimation nodes. As these two nodes only communicate with each other during the consensus operation, the degree is $\phi_i = \phi_j = 1$.

Weights π_{ij} can be arranged into a transition matrix $\Pi \in \mathbb{R}^{n_{est} \times n_{est}}$; according to the definition of the weights (49), the matrix can be calculated as:

$$\Pi = \begin{bmatrix} \pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22} \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \tag{50}$$

This matrix is symmetric and doubly stochastic, which is the condition to ensure that an average consensus between nodes is reached after a number of iterations. In order to ensure such convergence, the consensus step is repeated for $\nu = 1, \ldots, 5$ at each time step k.

IV. Victim position estimation algorithm

The estimation of the victim's position is performed once at least 3 rescuers enter the distance estimation zone. This step relies on a *WLS* trilateration algorithm together with *GDOP* considerations to improve accuracy and assess estimation quality.

A. Victim trilateration algorithm

This algorithm shares its mathematical principles with the trilateration process described in Section (III-B). However, as the victim's position remains unchanged, no Kalman filter has been implemented.

Without considering the preliminary step described in Section (III-A), the same structure of equations (23)-(24)-(25)-(26) has been utilized again to construct new H and z matrices, considering:

- $x = [x_v \quad y_v]^\top$ as the unknown position of the victim
- d_i as the estimated distance between the i^{th} rescuer and the victim, obtained through the strength of the magnetic field detected by the *ARTVa* sensor.
  Assuming a $2D$ field (as described in (2)), the magnetic field decays proportionally to $1/d^2$. Thus, d has been calculated as:

  $$d = d_{\text{ref}} \left(\frac{B_{\text{ref}}}{B} \right)^{\frac{1}{2}} \tag{51}$$

  where B is the detected field intensity and B_{ref} is a reference value corresponding to the distance d_{ref} at which phase 4 of the *SAR* operation begins (according to Section (I-A)).
- $[x_i \quad y_i]^\top$ as the estimated position of the i^{th} rescuer (with associated position covariance P_i), coming from the rescuers' position estimation algorithm (Section III).

At each time step k new measurements are performed. This process adds a new row to the matrices H and z, resulting in a growing system of linearized equations.

$$H = \begin{bmatrix} H_{k-1} \\ H_k \end{bmatrix} \quad \text{and} \quad z = \begin{bmatrix} z_{k-1} \\ z_k \end{bmatrix} \tag{52}$$

B. WGDOP and WLS algorithms

The overall uncertainty in the estimation of the victim's position is influenced by two main factors:

1) The variability of the estimated distance between each rescuer and the victim.
2) The spatial configuration of the rescuers around the victim.

Both effects have been considered in the estimation process by combining *WLS* and *WGDOP* methods.

1) *Distance uncertainty*

The confidence in the estimated distance d_i depends on the strength of the detected magnetic field, which increases the closer the *ARTVa* sensor is to the victim beacon [3] [4]. An exponential relationship has been considered:

$$\sigma_{\text{ARTVa},i} = \sigma_{\min} + (\sigma_{\max} - \sigma_{\min}) \cdot \exp\left(-3 \cdot \frac{B - B_{\text{victim}}}{B_{\text{ref}} - B_{\text{victim}}} \right) \tag{53}$$

where:

- $\sigma_{\min}$ and $\sigma_{\max}$ represent the minimum and maximum uncertainty values.
- B_{victim} and B_{ref} represent the minimum and maximum magnetic field values.

The $\sigma_{\text{ARTVa},i}$ values have been used to compute a measurement covariance matrix C following the same structure described by equations (27)-(28)-(29)-(31)-(32). As per H and z, matrix C also grows with each new measurement, according to a block diagonal scheme:

$$C = \begin{bmatrix} C_{k-1} & 0 \\ 0 & C_k \end{bmatrix} \tag{54}$$

A weight matrix W has been then defined as follows:

$$W = C^{-1} \tag{55}$$

This weight has been used in the final victim position estimation $\hat{x}$, following a standard *WLS* approach:

$$\hat{x} = \begin{bmatrix} \hat{x}_v \\ \hat{y}_v \end{bmatrix} = \left(H^\top W H \right)^{-1} H^\top W z \tag{56}$$

2) *Geometric uncertainty*

The geometric configuration of the rescuers affects the confidence in the trilateration process. This has been quantified using the Weighted Geometric Dilution of Precision (*WGDOP*) [11] [12], which captures the combined impact of geometry as well as the varying measurement confidence levels.

$$WGDOP = \sqrt{\text{Trace}\left((H^\top W H)^{-1} \right)} \tag{57}$$

A lower *WGDOP* value indicates a more favorable geometry for localization (i.e. rescuers are well distributed around the victim) and lower expected uncertainty in the position estimate, while high *WGDOP* values imply poor estimation accuracy due to far away, nearly collinear or clustered rescuers' positions.

The *WGDOP* value has not been used directly in the estimation algorithm; however, its value serves as an indicator of the quality of the geometry and of the accuracy of the dynamic anchor geometry during the search operation.

V. Simulation results

In order to provide a better understanding of the *MATLAB* simulation results and verify that all algorithms work as intended, the outputs of a single representative run are presented below. Then, given that many simulation parameters are stochastic, a batch of multiple runs has been performed under different conditions to conduct a more comprehensive statistical evaluation and comparative analysis.

A. Single simulation results

The detailed behavior observed in a typical single simulation run has been reported below, highlighting the dynamics of the system and the evolution of the estimation metrics. A total number of rescuers $n = 5$ has been considered.

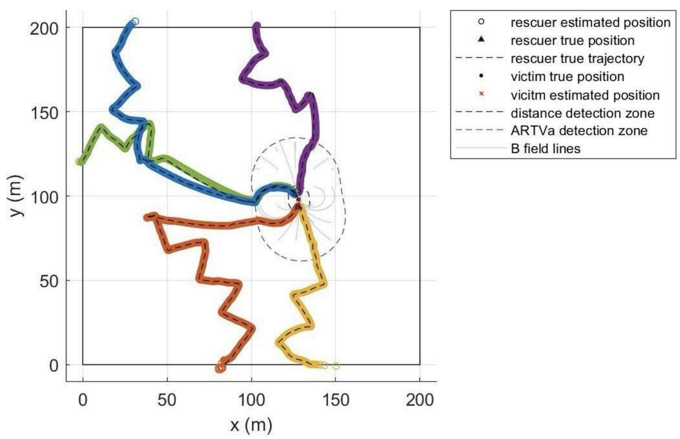

Fig. 4. Single simulation dynamic plot

Two primary metrics have been plotted:

- the trace of the covariance matrix (Fig. 5), which represents the total variance of the estimated state across all variables and returns the confidence the estimated positions over time
- the residuals between the true and the estimated positions of each rescuer (equal to the norm of the estimation error, Fig. 6), which provide a measurement of the estimation accuracy.

$$\text{res}_i = ||x_{\text{rescuer i, real}} - x_{\text{rescuer i, estimated}}|| \quad (58)$$

By observing these 2 metrics, it's evident how the rescuer position estimation algorithm performs well, as both the absolute position error and the estimation uncertainty quickly converge to low values early in the simulation. The initial values are high as a very large initial covariance has been selected to simulate the complete lack of prior knowledge regarding the relative positions of the rescuers. However, they tend to rise by a small amount later in the simulation for some rescuers. This behavior can be explained by considering how the algorithm has been built: the relative positions of the rescuers play a big role in the estimation process. As described in Section (III-A) indexing is circular; if, considering the i^{th} rescuer, the $(i+1)^{\text{th}}$, $(i-1)^{\text{th}}$ and $(i+2)^{\text{th}}$ anchors are clustered together and far from the estimated position of i, then the estimation

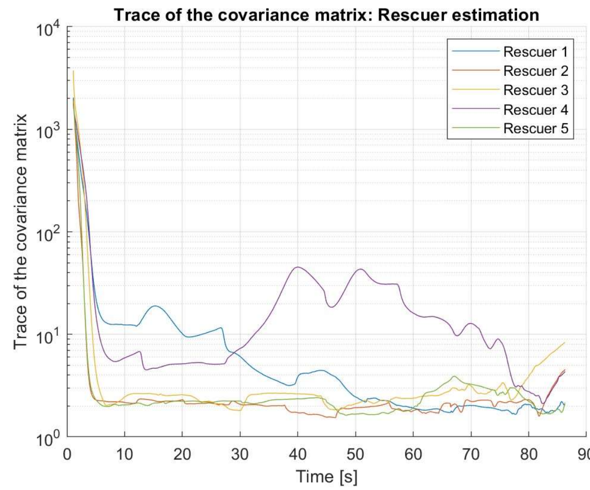

Fig. 5. Trace of the covariance matrix over t for every rescuer

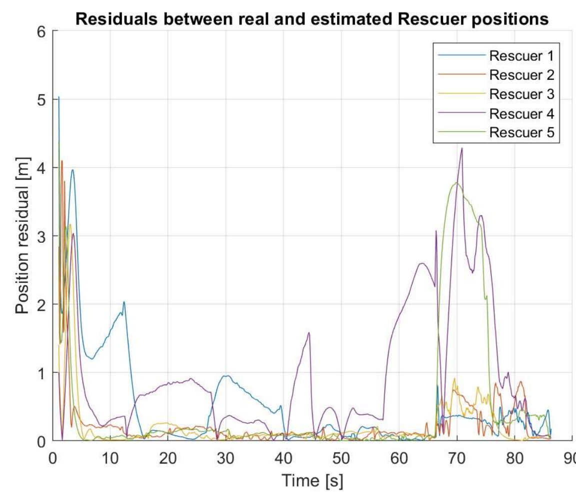

Fig. 6. Residuals over t for every rescuer

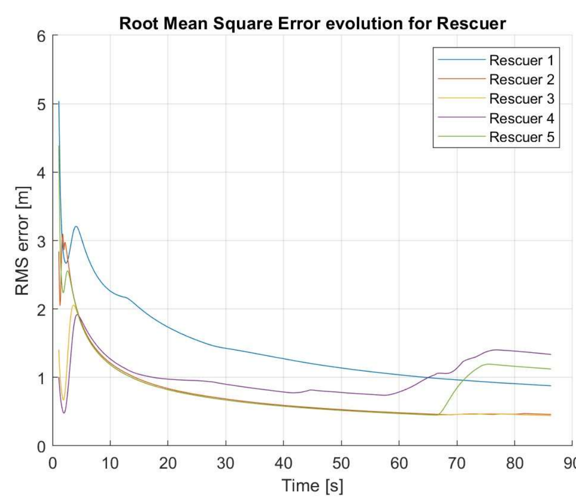

Fig. 7. Root Mean Square error over t for every rescuer

quality degrades. This situation is more likely towards the end of the simulation when the closest rescuers have already reached the estimated position of the victim while others are still converging, leading to a temporary rise in the estimation error parameters.
This behavior is therefore consistent with the structure of the simulation; it should not be interpreted as a failure of the algorithm, but as the consequence of the dynamics of the system. Its impact is less pronounced when considering the Root Mean Square (*RMS*) error (Fig. 7), which accounts for the evolution over time of the residuals of the estimation process (rather than focusing only on instantaneous variations): the spike is less pronounced, confirming the reliability of the process.

$$RMS_t = \sqrt{\frac{1}{t}\sum_{k=1}^{t}\text{res}_k^2} \tag{59}$$

As far as the victim position estimation is concerned, similar parameters have been analyzed. In this case, the estimation refers to a single target, so only one value is tracked.
By plotting the trace of the covariance matrix (Fig. 8), it can be observed again how it quickly drops to near-zero levels, indicating the high level of confidence in the estimation once all rescuers approach the victim's position. Residual analysis

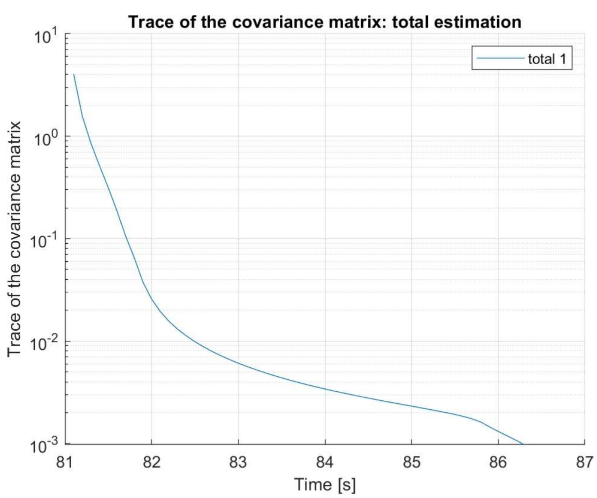

Fig. 8. Trace of the covariance matrix over t for the victim estimation

(Fig. 9) also conveys the idea of a robust estimation. As per the rescuer position estimation, some late-stage rising in the instantaneous residual might happen as a rescuer that was still outside the victim estimation circle enters it and disrupts the overall estimation. However, because of the weighting matrix W, this disruption only has a marginal impact on the accuracy of the global estimation. The evolution of the *WGDOP* (Fig. 10) has also been considered. Its value decreases rapidly to values close to zero as rescuers converge towards the victim's position. This behavior confirms the effectiveness of the system's geometry during the estimation process: as the spatial distribution of the rescuers improves around the

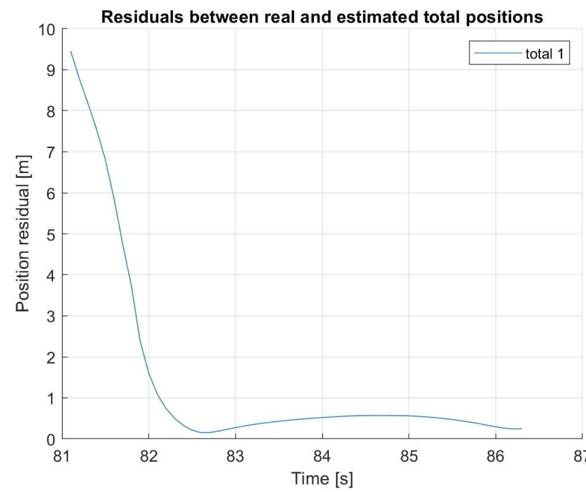

Fig. 9. Residual over t for the victim estimation

victim, the dilution of precision decreases, resulting in a more accurate victim position estimate. Although new rescuers entering the distance estimation circle cause a disturbance, the weighting provided by matrix W mitigates these effects, ensuring robustness.

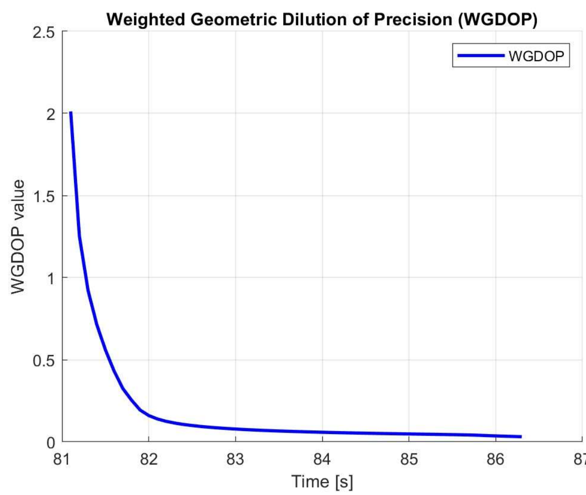

Fig. 10. *WGDOP* over t

B. Multiple simulation results

A total of 180 different simulations considering a different number of total rescuers has been performed to validate the proposed algorithm:

- 60 instances considering $n = 3$
- 60 instances considering $n = 4$
- 60 instances considering $n = 5$

a smaller number of rescuers has not been simulated, as at least 3 are needed to perform the victim estimation process. A higher number has also not been considered, as groups larger than 5 or 6 people tend to be uncommon.

Given that the duration of each simulation varies, the time-axis has been normalized in order to perform an adequate comparison across different scenarios. To analyze quantities involving multiple rescuers from different simulations (like those at Fig. 5, 6 and 7), the average of the results coming from the different rescuers has been considered. Then, to obtain a representative trend, the average across all simulations has been calculated.

The average of the trace of the covariance matrix for the rescuer position estimation process can be visualized at Fig. 11, comparing scenarios with varying numbers of rescuers.

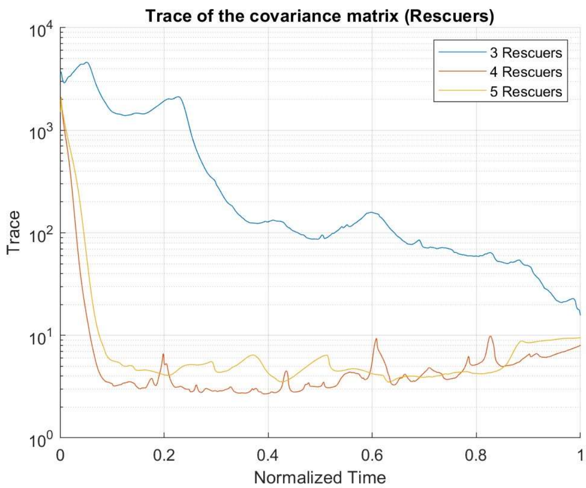

Fig. 11. Trace of the covariance matrix over normalized t for the rescuer estimation process over multiple simulations

It can be observed that for the simulations having $n = 4$ and $n = 5$, the average of the variance is almost identical. As described in (V-A), in the late stages of the *SAR* operation, the spatial configuration of the entities may be suboptimal, leading to a slight rise in the overall covariance. In contrast, simulations with 3 rescuers show an overall higher covariance in the whole operation, reaching comparable levels only in the last instances of the simulation, when all rescuers are very close to the victim's source of the magnetic field. This is to be attributed to the fact that each rescuer is unable to define two distinct sets of anchors for position triangulation (as described in (III-A)), thus performing a consensus operation using two identical measurements.

The same conclusions can be drawn by looking at the average of the residuals (Fig. 12) and the average of the *RMS* (Fig. 13) across simulations. Once again, results for $n = 4$ and $n = 5$ are almost identical, demonstrating the accuracy of the proposed algorithm in different scenarios.

Considering the victim estimation process, much of the same considerations can be made. The evolution of the average across simulations of the trace of the covariance matrix (Fig. 14) confirms the increase in the estimation precision as the rescuers converge towards the victim. Given that the victim trilateration process is based on the rescuer estimated positions

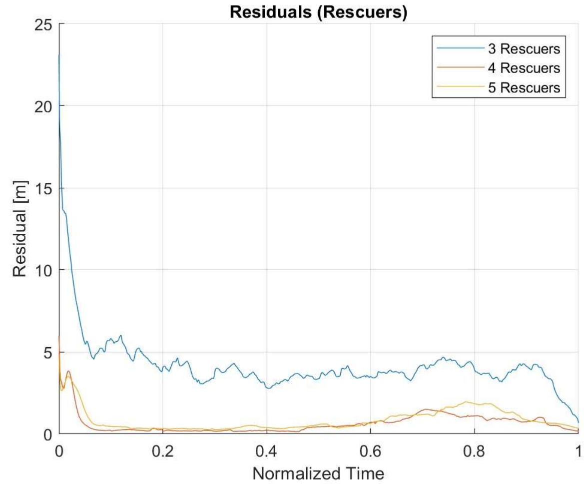

Fig. 12. Residuals over normalized t for the rescuer estimation process over multiple simulations

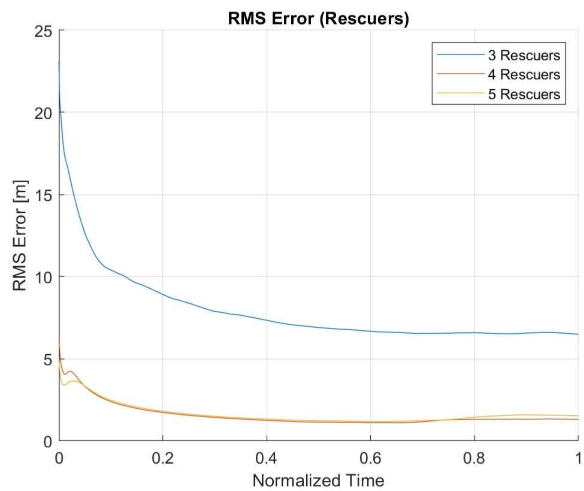

Fig. 13. Root Mean Square error over normalized t for the rescuer estimation process over multiple simulations

(Section (IV)), $n = 4$ and $n = 5$ simulations have nearly identical performance, highlighting how increasing the number of rescuers does not lead to significant improvements in estimation quality. $n = 3$ cases still suffer from poor positional performance, leading to a less confident victim estimation.

The average of the residuals of the victim's position estimation (Fig. 15) confirms this trend: the estimation error decreases as rescuers approach the victim, reaching comparable levels across all scenarios in the final stages.

The evolution of the average of the *WGDOP* index across simulations (Fig. 16) further highlights the improvement in positional accuracy as rescuers close in on the victim. Another result to be highlighted is that the initial *WGDOP* values are lower in $n = 3$ simulations, gradually increasing in scenarios with $n = 4$ and $n = 5$. This behavior suggests that

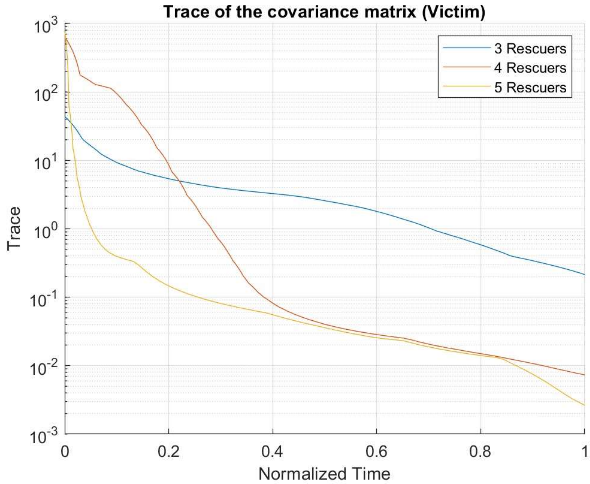

Fig. 14. Trace of the covariance matrix over normalized t for the victim estimation process over multiple simulations

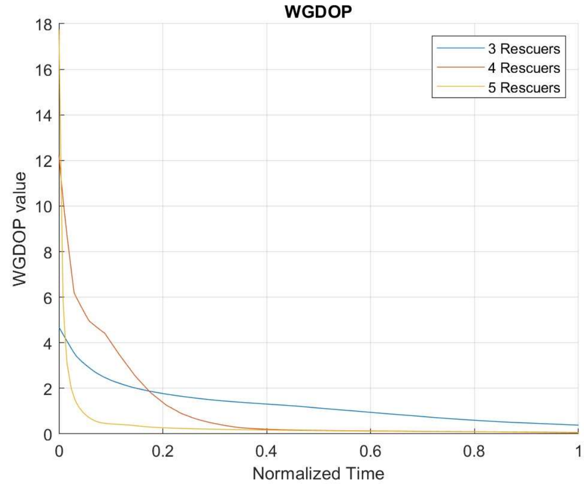

Fig. 16. *WGDOP* over normalized t across multiple simulations

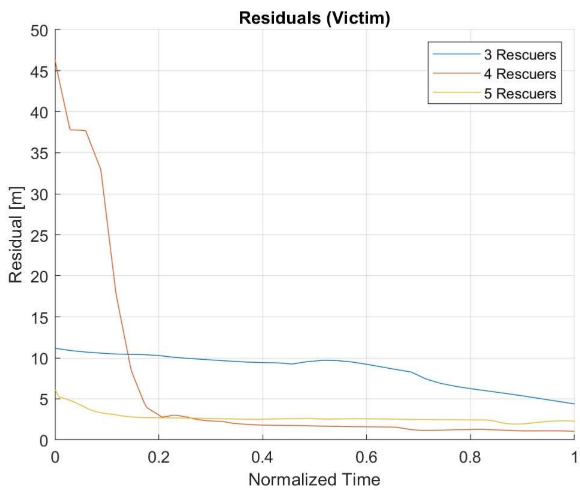

Fig. 15. Residuals over normalized t for the victim estimation process over multiple simulations

smaller teams benefit from a statistically better initial spatial distribution, providing better geometric conditions in the early stages of the localization.

VI. Conclusions

In this work, a distributed avalanche *SAR* algorithm has been developed and validated. The method proposes a cooperative strategy among a team of rescuers, each equipped with an *ARTVa* receiver and 2 *UWB* devices, with the objective of jointly estimating both their positions relative to one another and the position of buried victim.

The algorithm is based on recursive *UKF* and consensus strategies, allowing the rescuers to progressively refine their estimates of both their own state as well as the victim's position.

The algorithm has been tested by performing a total of 180 simulations across different team configurations. The results were averaged and normalized over time, extracting representative trends.

The final results confirm the effectiveness of the proposed approach:

- The rescuers' position estimation benefits from cooperative behavior. Simulations with $n = 4$ and $n = 5$ rescuers show almost identical performance; meanwhile, in simulations with $n = 3$, consensus cannot be performed and results are worse.
- The victim position estimation improves as rescuers approach the target.
- The analysis of the *WGDOP* index confirms the geometrical improvement during search.

Overall, the proposed approach is robust and scalable under the tested conditions. Future developments could include the modeling of environmental factors such as an irregular and sloped working area, terrain obstacles and signal interferences; also, additional sensors may be integrated to improve robustness in less-than-ideal environments. Lastly, $3D$ simulations accounting for multiple victims and burial depth would extend the applicability of the algorithm.

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